Comparing Air Pollution Trends Between Delhi, India and Dhaka, Bangladesh from 2003-2023

ENV 872L Final Project

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Introduction

Air pollution is one of the most pressing environmental and public health challenges of the 21st century, particularly across South Asia, where rapid urbanization, industrial growth, vehicular emissions, and agricultural waste burning have made several cities among the most polluted in the world (Ditta et al., 2026). Delhi, the capital of India, and Dhaka, the capital of Bangladesh, are two such megacities that consistently rank among the worst for air quality globally, with both cities experiencing severe seasonal pollution cycles driven by winter temperature inversions, stubble burning, brick kilns, and vehicle emissions along the shared Indo-Gangetic Plain (Ditta et al., 2026; Begum et al., 2018). Despite their geographic proximity and shared pollution sources, the two cities have followed divergent regulatory paths. India launched its National Clean Air Programme in 2019 targeting significant reductions in particulate matter (Garg et al., 2022), while Bangladesh was among the first South Asian nations to ban two-stroke engine vehicles in the early 2000s and transition to CNG-powered alternatives (State of Global Air, n.d.), raising important questions about whether these differing policy trajectories have translated into meaningfully different pollution trends over time.

While prior studies have individually examined air quality trends in Delhi (Garg et al., 2022; Singh et al., 2023) and Dhaka (Islam et al., 2021; Begum & Hopke, 2018), long-term comparative analyses spanning both cities remain limited. This project addresses that gap by using 20 years of monthly air quality data (2003 to 2023) to compare air quality trends in PM2.5, PM10, ozone (O3), and AQI across both cities. This study addresses the following hypotheses:

H_1: Air quality in Delhi and Dhaka exhibits a significant seasonal pattern over the 2003–2023 period.

H_2: Air quality trends in Delhi and Dhaka are significantly different over the 2003–2023 study period.

H_3: Mean air pollution levels differ significantly between Delhi and Dhaka across pollutants (PM2.5, PM10, O3, and AQI).

Data description

Dhaka Dataset

The Dhaka air quality data was sourced from the Bangladesh Air Quality Index (AQI) Dataset (2000–2025), published by Tapon Paul of Daffodil International University and retrieved from Mendeley Data (Paul, 2026). The full dataset contains 1,048,551 hourly records spanning 103 cities across Bangladesh across 13 columns: city_id, city_name, lat, lon, datetime, pm10, pm2_5, carbon_monoxide, carbon_dioxide, nitrogen_dioxide, sulphur_dioxide, ozone, and aqi. AQI values are calculated according to the US EPA standard.

Delhi Dataset

The Delhi air quality data was sourced from the Processed Delhi Air Quality and Weather Data dataset curated by Abhinav Sarkar and retrieved from Hugging Face (Sarkar, 2024). The full dataset contains approximately 2.9 million hourly observations from multiple monitoring locations across Delhi from March 2000 to November 2024, across 15 columns: city, event_timestamp, temperature, humidity, pressure, wind_speed, wind_direction, pm25, pm10, no2, so2, o3, co, aqi, and location_id. The dataset was pre-processed prior to publication, with missing AQI values removed and remaining gaps addressed through mean imputation and nearest neighbor interpolation.

Methods

Initial Data Wrangling

Both datasets were cleaned and processed separately before being combined for analysis. The Dhaka dataset was filtered to retain only observations from Dhaka city. For both datasets, the datetime columns were

converted into a proper R datetime object using the ymd_hms function and only the columns needed for analysis were selected: Location, Month, Year, PM2.5, PM10, O3, and AQI. Both datasets were filtered to January 1, 2003 through December 31, 2023. We chose this window because the Dhaka dataset spans a longer date range from 2000 to 2025 and the Delhi dataset does not have complete monthly records for 2000 and 2024. To keep the data consistent and comparable across both cities, we focused on a clean 20-year trend. Hourly observations for both datasets were aggregated into monthly averages for all four pollutants (AQI, PM 2.5, PM 10, and O3) using na.rm = TRUE to handle any missing values. A Location variable was added to each dataset before using rbind to combine the two datasets. A date column marking the first day of each month was created using make_date for use in visualizations and regression analyses. The final combined dataset contained 504 average monthly observations over 8 columns (see below).

```
## 'data.frame':    504 obs. of  8 variables:
##  $ month      : int  1 1 1 1 1 1 1 1 1 1 ...
##  $ year       : int  2003 2004 2005 2006 2007 2008 2009 2010 2011 2012 ...
##  $ PM2.5      : num  70 74.6 82.5 88.9 95.5 ...
##  $ PM10       : num  119 127 139 151 162 ...
##  $ O3         : num  42.4 42.8 43.2 42.5 42.6 ...
##  $ AQI        : num  151 156 162 166 170 ...
##  $ Location   : chr  "Dhaka" "Dhaka" "Dhaka" "Dhaka" ...
##  $ first.month: Date, format: "2003-01-01" "2004-01-01" ...
```

Visualization

To get an initial sense of how pollution has changed over time in each city, we plotted monthly trends for all four pollutants. Both cities were displayed on the same graph with a linear trend line to highlight the overall direction of change across the 20-year study period.

Time Series

We built monthly time series objects for each pollutant and each city. Each series was then decomposed into the seasonal, long-term trend, and remainder components.

Trend Analysis

To test whether pollution levels were changing significantly over time, we ran both seasonal and non-seasonal Mann-Kendall tests for each pollutant in each city. We chose to run both because the initial graphs suggested a more likely seasonal trend in Dhaka's air quality with weaker seasonality in Delhi's air quality. The non-seasonal Mann-Kendall test was run on the de-seasonalized series, obtained by subtracting the seasonal component from the STL decomposition.

Comparative Analysis

Before running any comparative analyses, we had to check if our data was normally distributed. For this, we generated some qq plots for all four variables for each city. Then, to compare the air quality between the two cities, we used two approaches. First, we ran two-sample t-tests for each pollutant to test whether mean pollution levels over the full study period were significantly different between Delhi and Dhaka. Second, we fitted linear regression models with an interaction term between time and location for each pollutant, using the formula lm(pollutant ~ first.month * Location).

Results

Descriptive Trends

Monthly trends showed contrasting trajectories between Delhi and Dhaka. At the beginning of the study period, from 2003, Delhi had a substantially higher AQI and particulate pollution but showed a clear decline over time, after the mid 2000s. In contrast, Dhaka showed a gradual increase in AQI, PM 2.5, and PM 10, resulting in convergence in air quality values between the two cities in recent years (Figures 1-3). Ozone, however, followed a different pattern. Dhaka demonstrated a consistently higher concentration of ozone and a decline over time, whereas Delhi showed relatively stable ozone levels with a slight increase in recent years (Figure 4).

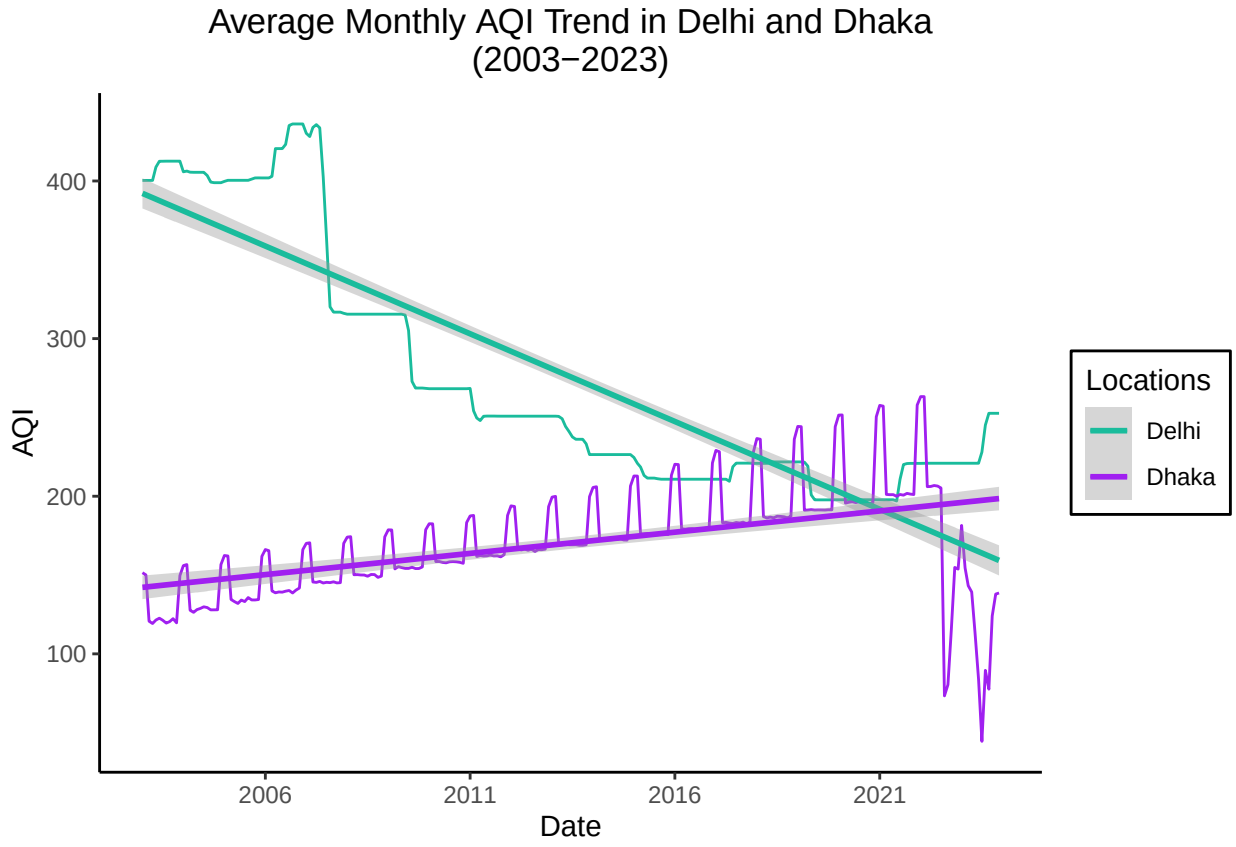

Figure 1: Monthly AQI Trend from 2003-2023

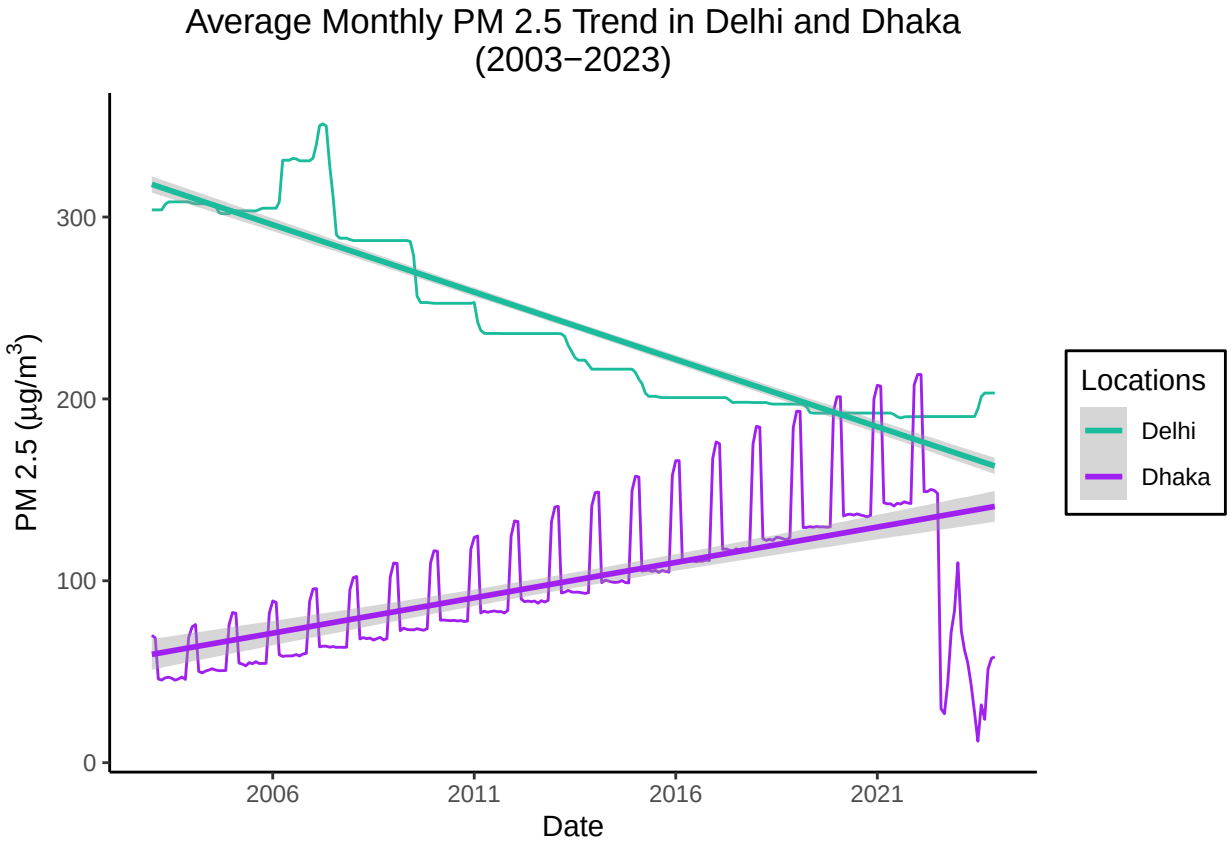

Figure 2: Monthly PM 2.5 Trend from 2003-2023

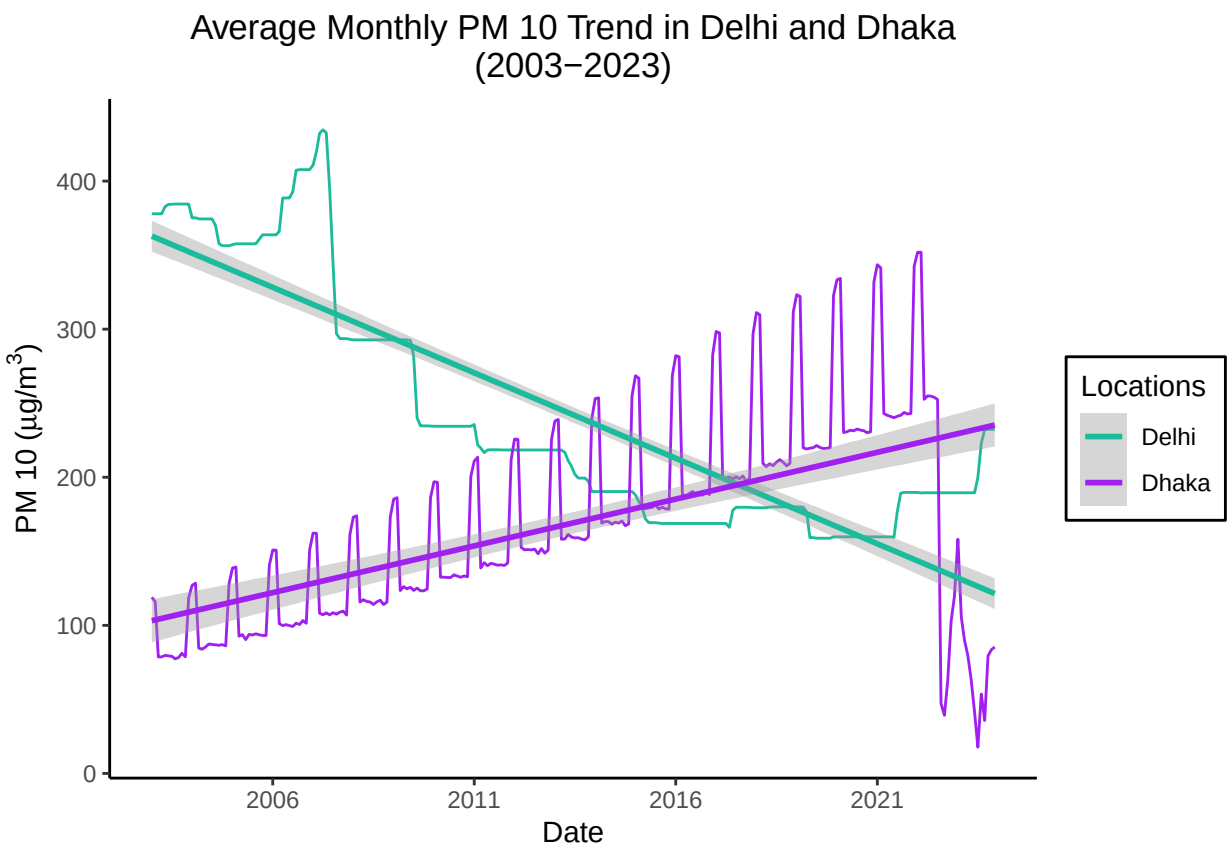

Figure 3: Monthly PM 10 Trend from 2003-2023

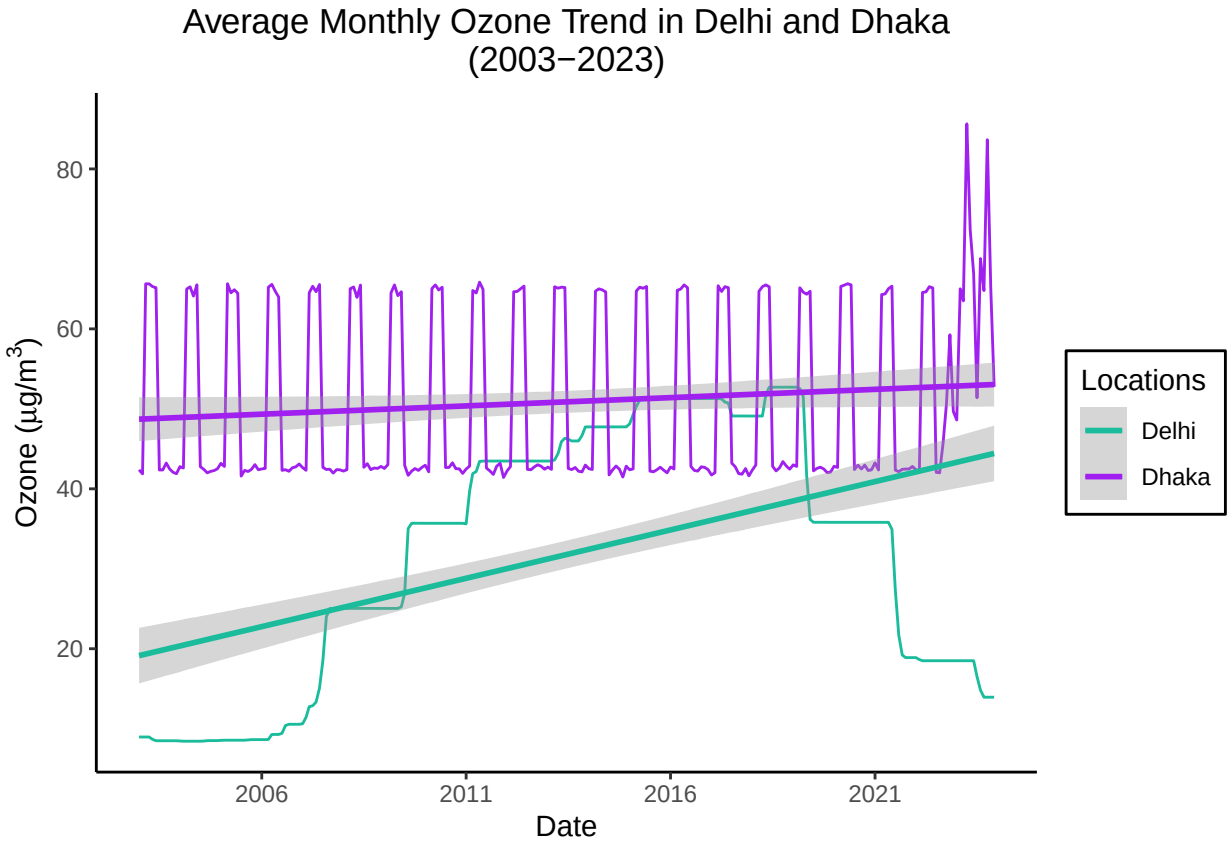

Figure 4: Monthly Ozone Trend from 2003-2023

Time Series Decomposition

The time series decomposition revealed a strong and consistent seasonality in both cities, with recurring peaks likely corresponding to winter pollution periods (Figures 5-6). Dhaka showed a more regular seasonal pattern while Delhi showed a much higher variability in the remainder component. The trend components broadly aligned with the patterns observed in the initial visualization, with declining particulate pollution in Delhi and increasing trends in Dhaka.

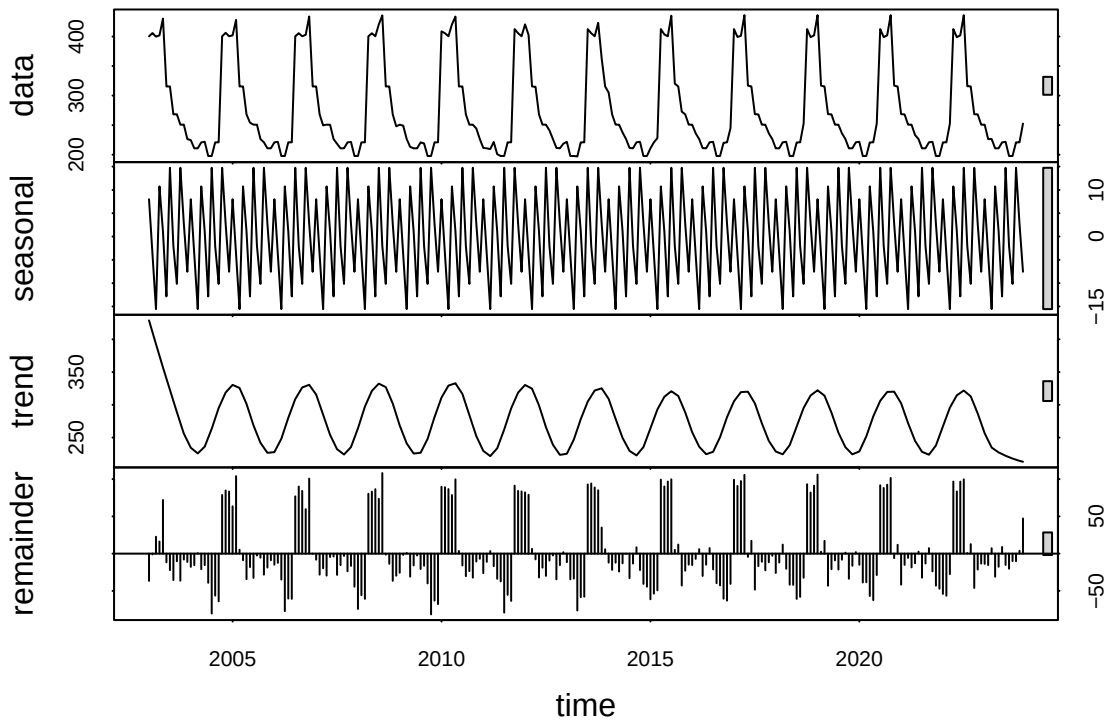

Figure 5: Decomposition of Delhi's AQI Time Series (2003-2023)

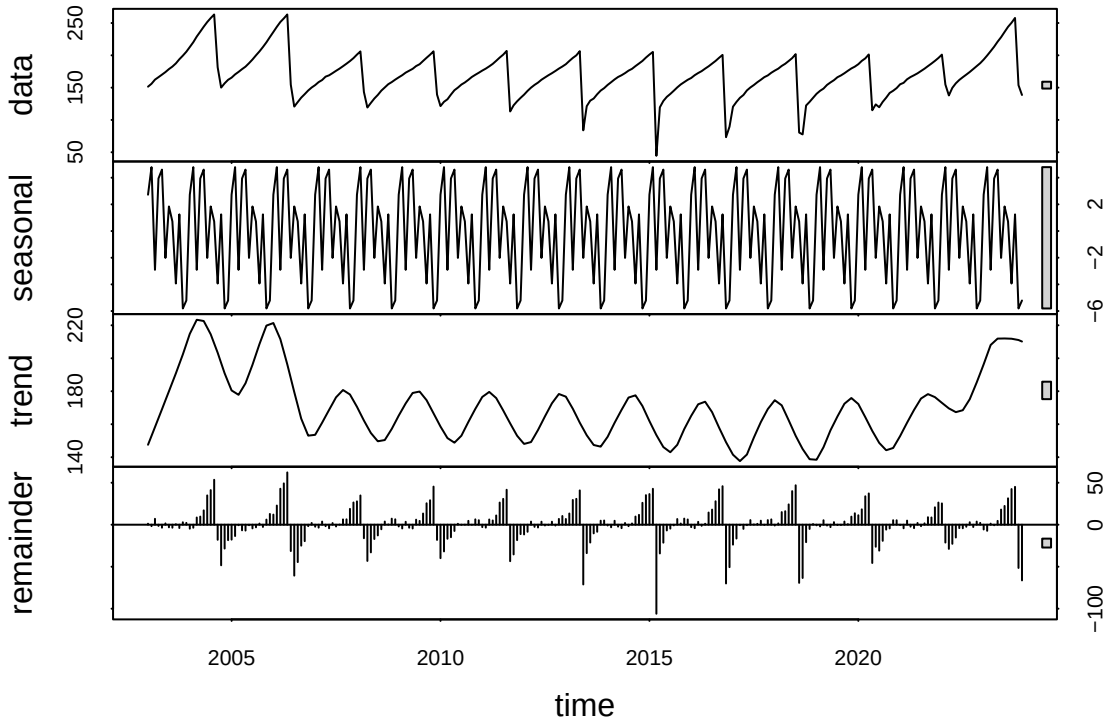

Figure 6: Decomposition of Dhaka's AQI Time Series (2003-2023)

Trend Analysis

The quantile–quantile plots showed that pollutant distributions deviate from normality in both cities, with stronger skewness and heavier tails observed in Delhi. The Mann-Kendall tests provided limited evidence of statistically significant trends (Table 1). Dhaka's ozone showed a strong decreasing trend ($p < 0.0001$) while Delhi's PM 2.5 showed to be significant only after the seasonal component was removed ($p = 0.033$). The other pollutant-location combinations were not statistically significant, indicating that the variability could have obscured any consistent long-term trends.

Table 1: Summary of Mann-Kendall Results

Location	Variable	Seasonal p-value	Non-Seasonal p-value	Trend Statistic
Dhaka	AQI	0.0768	0.2045	-204
Dhaka	PM 2.5	0.0768	0.2354	-204
Dhaka	PM 10	0.1068	0.2354	-186
Dhaka	O3	**1.43e-06**	**2.06e-06**	-554
Delhi	AQI	0.3226	0.1078	-114
Delhi	PM 2.5	0.0862	**0.0330**	-197
Delhi	PM 10	0.4257	0.1409	-92
Delhi	O3	0.3098	0.4421	117

Comparative Analysis

The two-sample t-tests showed significant differences in the mean pollution levels between the cities (Table 2). Delhi had a significantly higher mean AQI, PM 2.5, and PM 10, while Dhaka had significantly higher mean ozone concentrations (all $p < 0.001$).

Table 2: Summary of Two Sample T-test Results

Variable	p-value	Dhaka mean	Delhi mean
AQI	**2.2e-16**	170.35	275.71
PM2.5	**2.2e-16**	100.21	240.54
PM10	**2.2e-16**	169.18	242.17
O3	**2.2e-16**	50.87	31.78

Regression Analysis

The linear regression models showed that pollution trends differ significantly between Delhi and Dhaka for all pollutants ($p < 0.001$). For AQI, PM 2.5, and PM 10, Delhi showed a declining trend while Dhaka showed an increasing trend, consistent with the previously observed convergence over time. The model fit was strongest for PM 2.5 ($R^2 = 0.894$) and AQI ($R^2 = 0.812$), and weaker for ozone ($R^2 = 0.424$) (Table 3).

Table 3: Summary of Linear Regression Results

Variable	p-value	Min	Median	Max	R-squared
AQI	**2.2e-16**	-152.894	-1.296	93.331	0.812
PM2.5	**2.2e-16**	-127.439	-3.511	79.986	0.894
PM10	**2.2e-16**	-214.902	-5.708	128.747	0.634
O3	**2.2e-16**	-30.487	-4.640	32.740	0.424

Discussion

Hypothesis 1: Seasonal patterns and long-term trends

Time series decomposition clearly showed strong and consistent seasonal patterns in both cities, with pollution peaking during winter months and declining during the monsoon. This aligns with known meteorological drivers along the Indo-Gangetic Plain, such as winter temperature inversions and reduced atmospheric mixing which are known to trap pollutants near the surface, while monsoon rainfall improves atmospheric quality. Seasonal emission sources, including agricultural crop burning and festival-related activities such as Diwali, likely further contribute to elevated winter pollution levels.

Evidence for long-term trends is more mixed. While visual trends suggested declining particulate pollution in Delhi and increasing levels in Dhaka, Mann–Kendall tests detect statistically significant changes in only a subset of the pollutants. This suggests that high variability and strong seasonality may be obscuring consistent long-term trends, even when directional changes are visually apparent.

Hypothesis 2: Differences in air quality between cities

Delhi showed significantly higher mean AQI, PM2.5, and PM10 concentrations than Dhaka, confirming that particulate pollution has historically been more severe in Delhi. However, ozone showed the opposite pattern, with higher concentrations in Dhaka. This may be reflecting fundamental differences in how these pollutants are formed, as ozone is more of a secondary pollutant, driven by photochemical processes rather than direct emissions.

These findings highlight that overall air quality comparisons depend on pollutant type, and that framing one city as uniformly "more polluted" may be oversimplifying the underlying dynamics.

Hypothesis 3: Is Delhi's Air Quality Decreasing More Rapidly than Dhaka's?

Results from the regression analyses indicated that pollution trends differ significantly between the two cities. For AQI and particulate matter, Delhi showed declining trends while Dhaka showed increasing trends, resulting in convergence over time. This pattern was consistent across both visual and statistical analyses.

These opposing trajectories are consistent with differences in policy and development between the two cities. Delhi's decline may reflect the impact of emissions controls and air quality policies, while Dhaka's increases likely reflect rapid urbanization and industrial growth. However, these interpretations are associative rather than causal and should be taken with caution.

Limitations

This study had several limitations.

1. The time-series nature of the data violates the independence assumption of standard statistical tests, which may have affected the reliability of the p-values we obtained from the t-tests and linear regression models.
2. Aggregating hourly data into monthly averages may have also masked short-term pollution extremes and reduced variability.
3. Differences in data sources and preprocessing methods between Delhi and Dhaka, including the imputation in the Delhi dataset, affected comparability of the two datasets.
4. Simple linear regression models, as attempted here, do not account for seasonality or non-linear dynamics in pollution trends.

5. The irregular patterns observed in ozone data suggest that there may have been potential measurement inconsistencies that require cautious interpretation of our results.

6. Pollutant distributions were skewed, particularly in Delhi, which may have influenced the statistical estimates and increased sensitivity to extreme values.

Conclusion

This study demonstrates that while Delhi historically experienced substantially higher particulate pollution than Dhaka, its air quality has improved over time, whereas Dhaka's has worsened. As a result, the gap between the two cities has narrowed, with evidence of convergence in particulate pollution levels.

Ozone trends diverge from this pattern, with higher concentrations in Dhaka and differing temporal dynamics between the cities. Overall, these findings suggest that policy interventions in Delhi may be associated with improvements in air quality, while rapid urbanization in Dhaka has offset earlier gains.

Future work should incorporate more advanced time-series methods and additional cities to better understand regional pollution dynamics and strengthen causal inference.

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